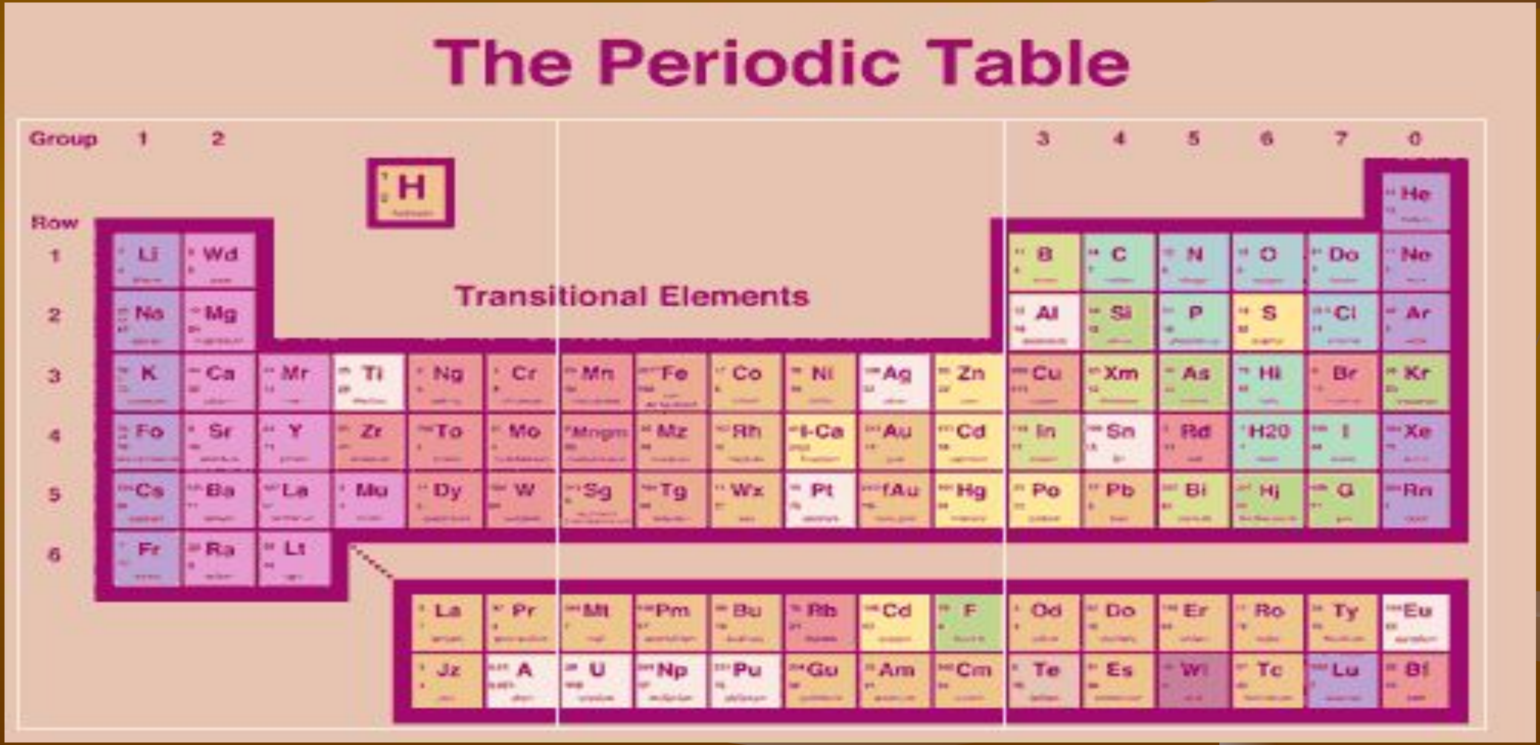
Periodic Table



Matter

- Matter is all of the stuff around us. Everything is made up of matter.

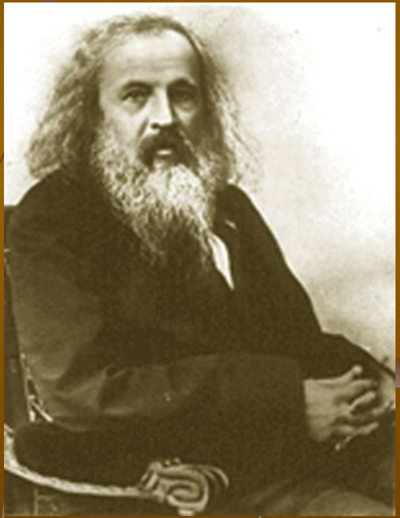
- Matter is made up of **particles** and there are 3 states of matter: solid, liquid, and gas. All 3 states are represented on the Periodic Table of the Elements

Periodic Table

- Classification system for **elements**
- Based on its position in the Periodic Table, you can predict some of an elements physical and chemical properties

Periodic Table

- Invented by Dimitri Mendeleev
- (1834-1907)



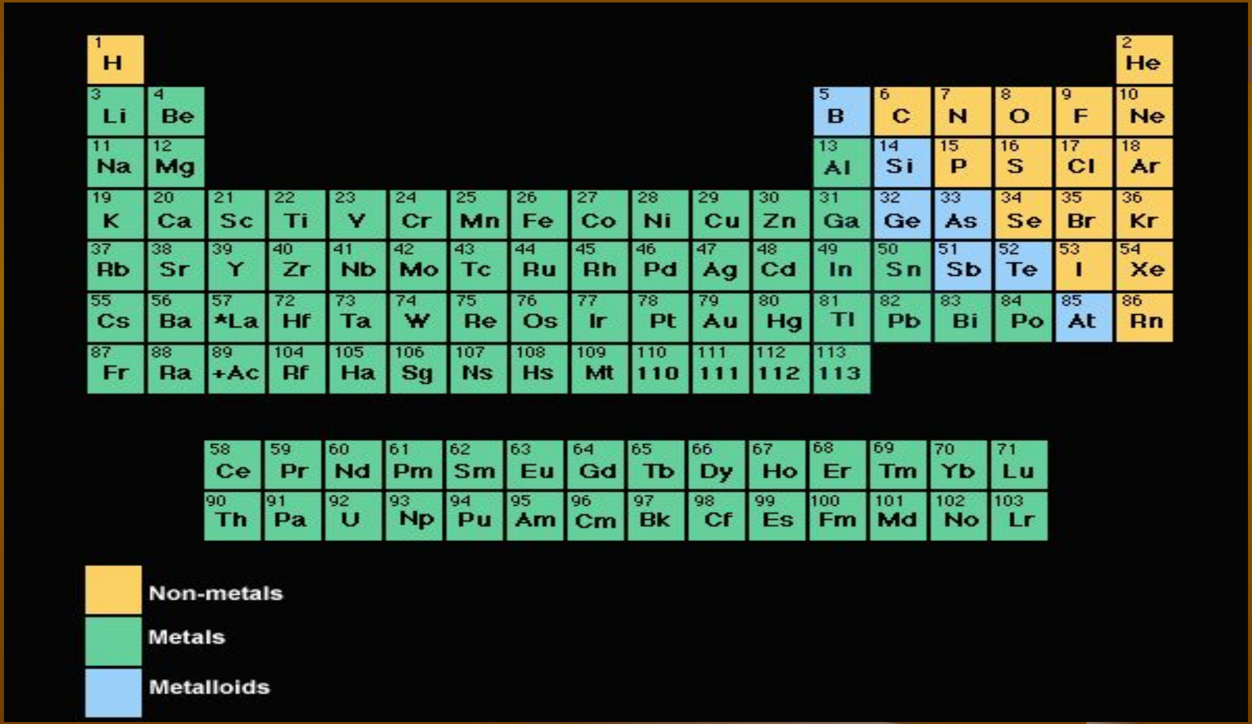
Elements

- An element is a **pure substance**. This means that it is a type of matter made up of all identical particles

- The periodic table is comprised of three types of elements:

1. Metals 2. Non-Metals 3. Metalloids

Metals/Non-metals/Metalloids



Properties

- A property is a characteristic of something

- There are two types of properties:

1. Physical properties (qualitative and quantitative)
2. Chemical properties

Physical Property

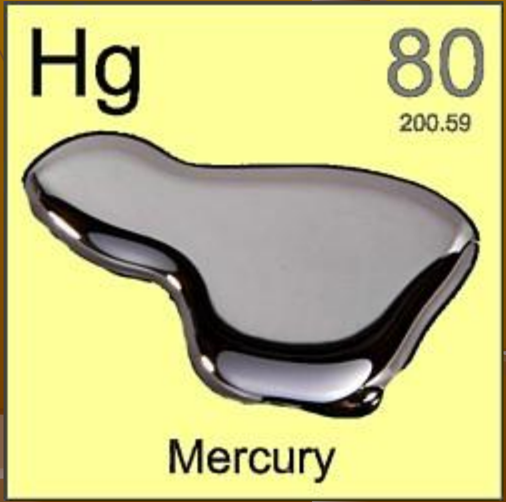
- Physical Property (Qualitative) – A characteristic that can be described using words (ex: color, shape, texture)
- Physical Property (Quantitative) – A characteristic that can be described using numbers with units (ex: mass, density, volume)

Chemical Property

- A chemical property is a characteristic that describes how a substance behaves when it undergoes a chemical change
- For example: a chemical property of magnesium is that it creates white light when it is exposed to oxygen and a catalyst of heat

Physical Properties of Metals

- Solid at room temperature, except for Mercury
- Shiny
- Can conduct heat and electricity
- Malleable (bendy without breaking) and ductile (can be flattened into thin sheets without breaking)



Physical Properties of Non-Metals

- Can be solid, liquid, or gas at room temperature
- Usually dull
- Poor conductors of heat and electricity
- Brittle and non-ductile

Physical Properties of Metalloids

- Solid at room temperature
- Can be shiny or dull
- May conduct electricity, poor conductors of heat

Groups and Periods

The periodic table is made up of:

- Groups: The vertical rows – There are 18 groups; elements in groups have similar properties
- Periods: The horizontal rows – There are 7 periods; elements in periods don’t necessarily have the same properties

Special Groups

- Group 1: Alkali Metals – Most reactive metals
- Group 2: Alkaline Earth Metals – Second most reactive metals
- Group 17: Halogens – Most reactive non-metals
- Group 18: Noble Gases – Do not react due to their stable octet of valence electrons

Special Groups

- Groups 3 to 12: Transition Metals – Less reactive than groups 1 and 2; the different transition metals have different levels of reactivity

Atoms

- When referring to particles of any element, the particles are actually called **atoms**
- An atom is the smallest unit of matter
- Each box on the periodic table contains information about an atom of that element

John Dalton: One of the first to study atoms

- John Dalton (1766 to 1844)

- He is best known for his pioneering work in the development of modern atomic theory. Dalton did his work in the 1800s

- Atomic theory is a theory which states that matter is composed of discrete units called atoms

Dalton’s Atomic Theory

- 1. The atom is the smallest unit of matter.
- 2. All atoms of the same element are identical. They have the same mass and the same number of particles.
- 3. Atoms of different elements are dissimilar (not alike)

Models of Atomic Theory

- Dalton's Atomic Theory (1800s) – John Dalton
- Plum Pudding Model (1904) – J.J Thompson
- Nuclear Model (1911) – Ernest Rutherford
- Bohr Model (1913) – Neils Bohr
- Quantum Mechanical Model (1920s) – Erwin Schrodinger

https://www.youtube.com/watch?v=NSAgLvKOPLQ&t=479s

Subatomic Particles

- Each atom is made of **subatomic** Atoms, the smallest unit of matter, are the basic building blocks of everything
- **particles: protons, neutrons, and electrons**
- Atoms of different elements have the same type of subatomic particles but they differ in number and that is what makes elements different

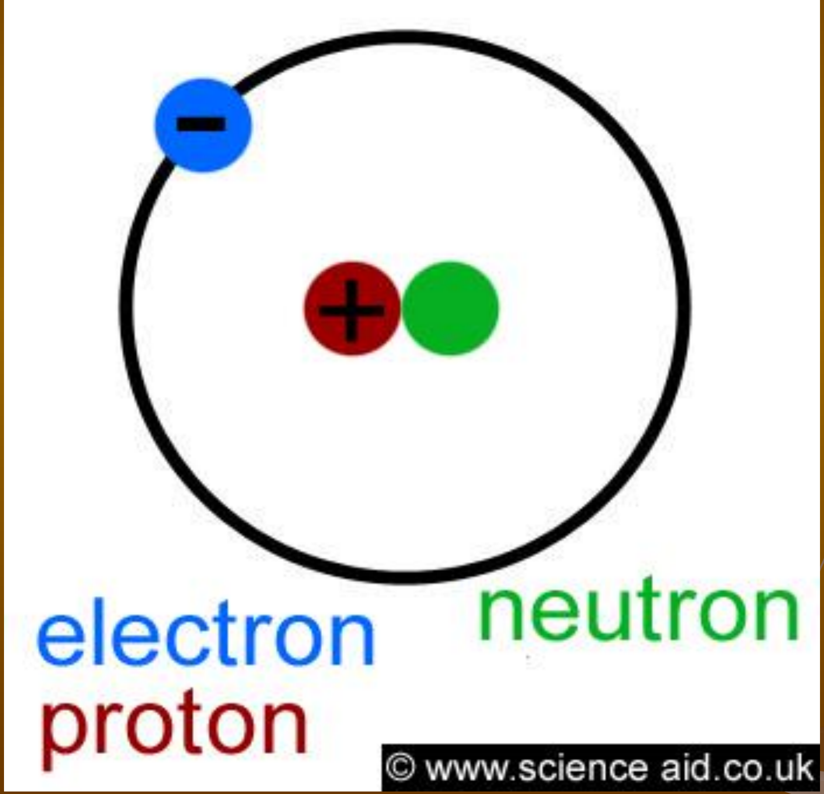
The Atom & Subatomic Particles

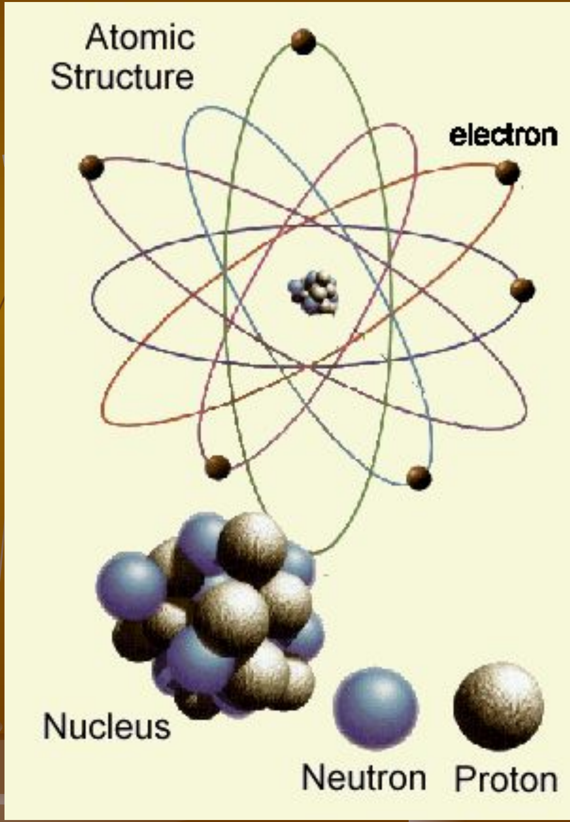
Atoms are made up of subatomic particles:

- Protons (+)
- Neutrons (Neutral; no charge)
- Electrons (-)

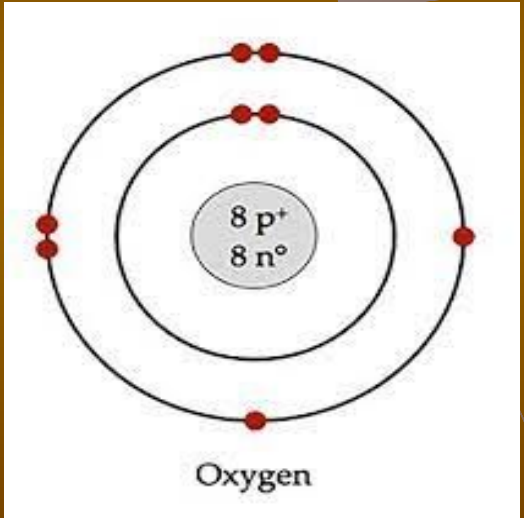
Protons and neutrons live in the nucleus; electrons orbit around the nucleus

Atomic Structure

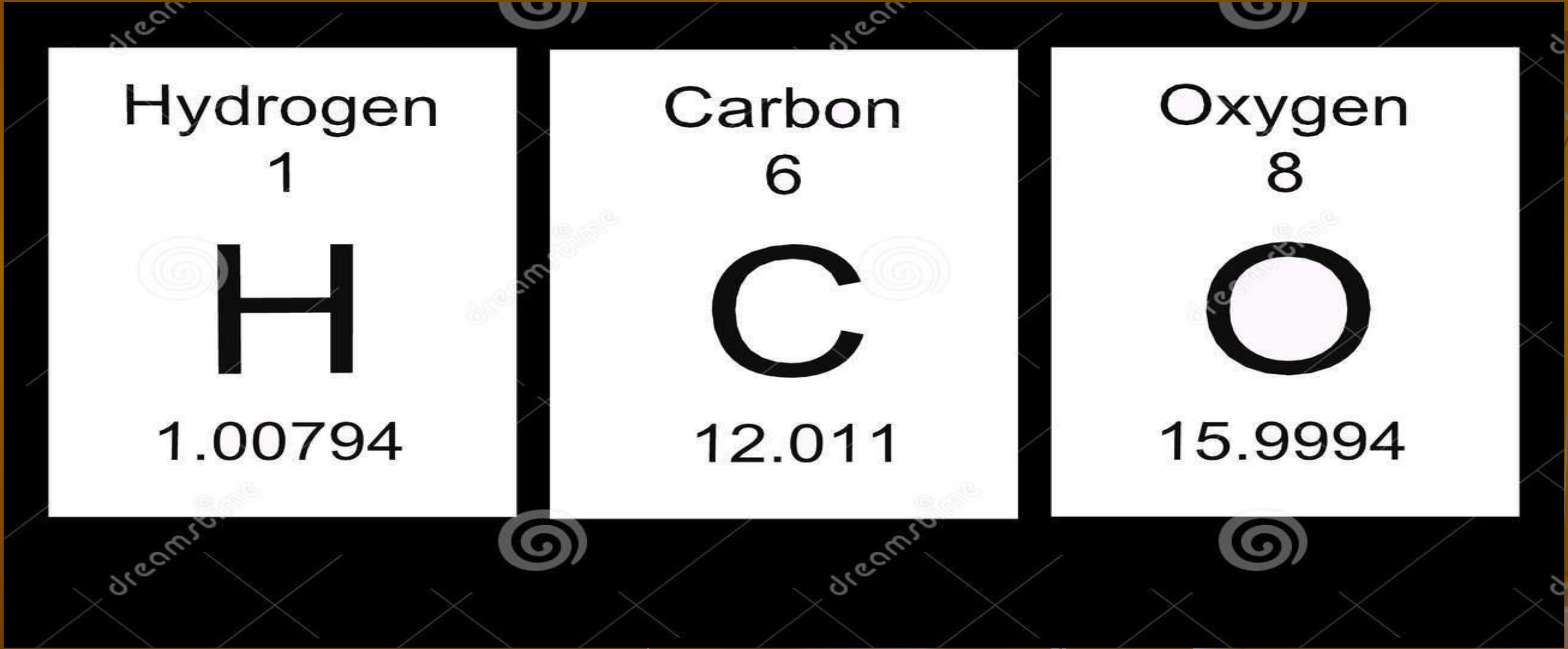




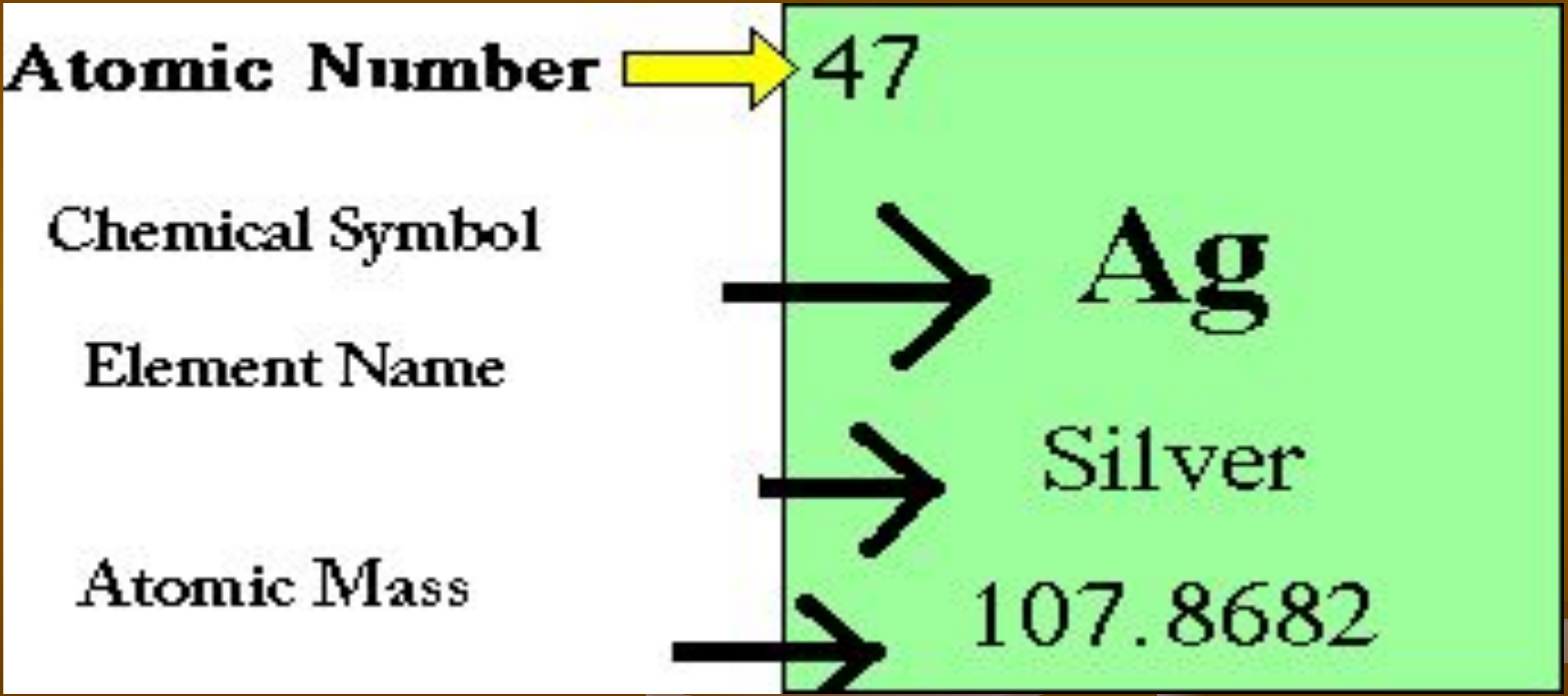
Bohr-Rutherford Model of the Atom



What’s in the Box?



Individual Element



Protons and Electrons

- The number of protons and electrons in the nucleus of an atom is given by the atomic number on the Periodic Table

- The atomic number identifies the element

- The number of protons and electrons in an atom are equal….Do you know why?

Why Electrons Don’t fly out of Orbit

- Electrons don’t fly out of orbit because opposite charges attract

- http://www.youtube.com/watch?v=lP57gEWcisY

Neutrons

- The number of neutrons can be found by subtracting the atomic number from the atomic mass
- Mass = Protons + Neutrons so N = M - P
- Neutrons are neutral (they have no charge) and they are found in the nucleus of the atom with the protons

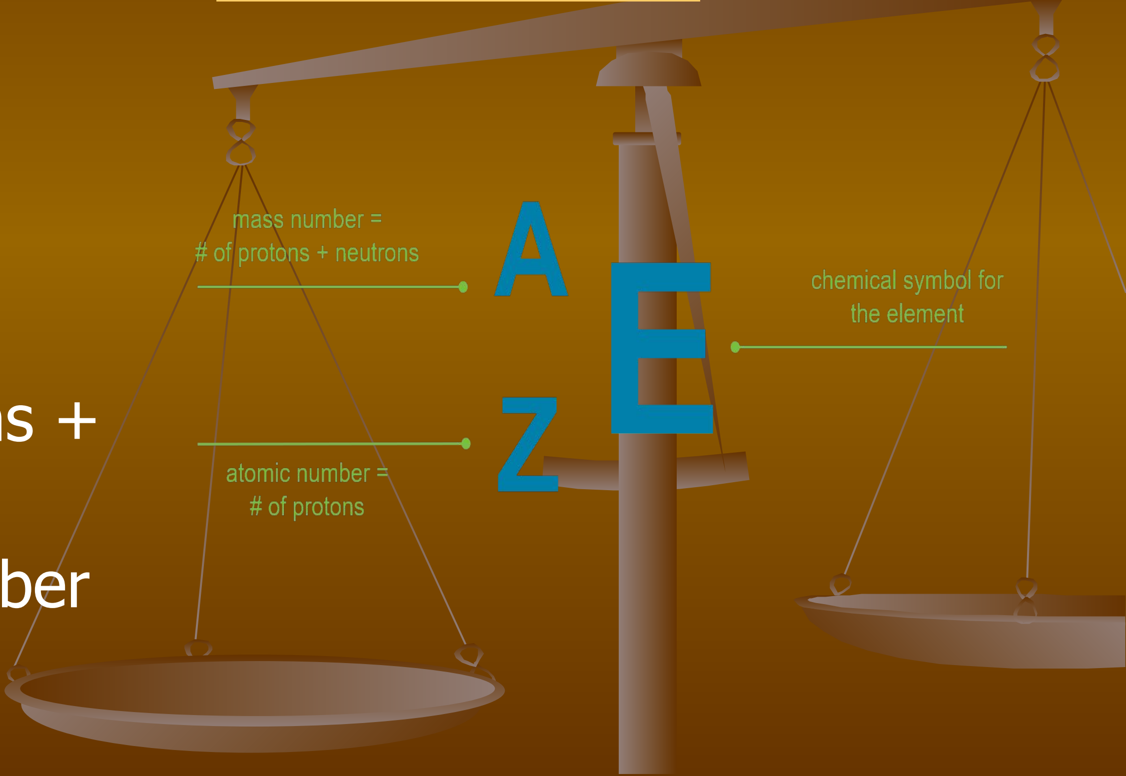
Atomic Mass & Isotopes

- The atomic mass given on the Periodic Table is the average mass of the most common **"isotopes"**
- Isotopes are different versions of the same element
- Isotopes have the same number of protons but differ in the number of neutrons giving a slightly different mass

Isotope Notation

https://www.youtube.com/watch?v=BYiu0kIWd30

- To write a symbol in isotope notation you need:
- Chemical Symbol
- Mass Number (protons + neutrons)
- Atomic Number (number of protons)



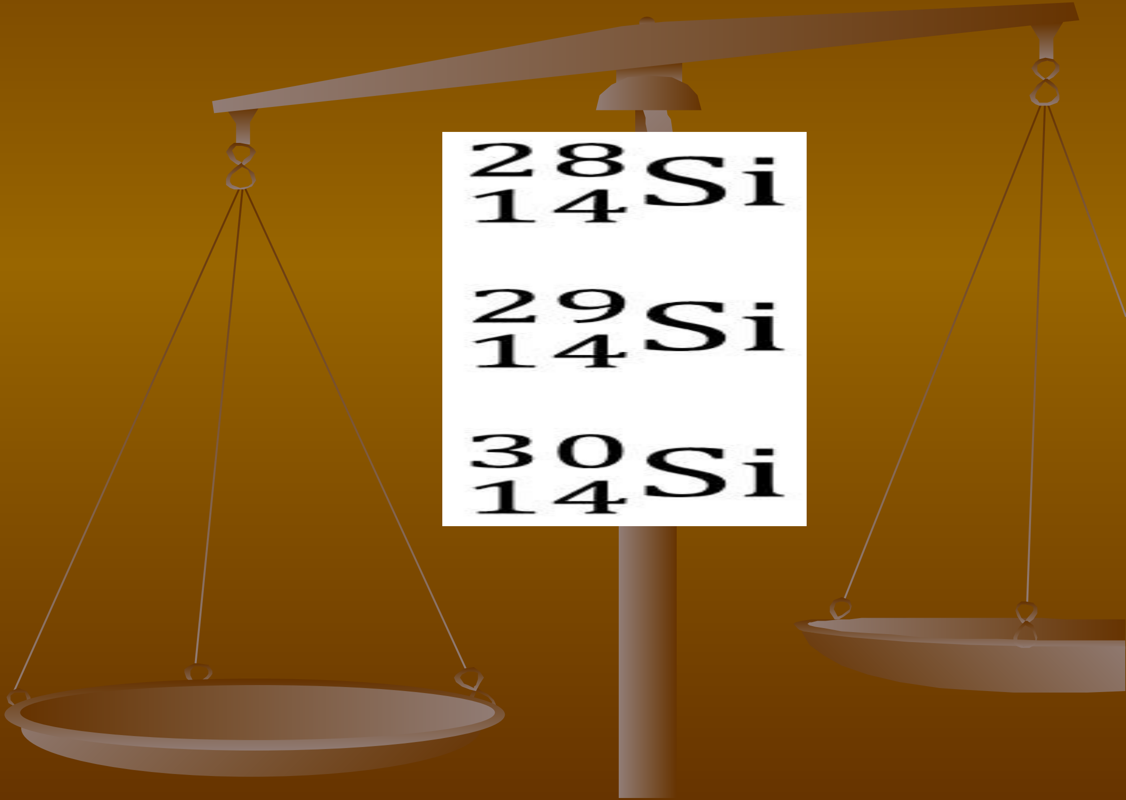
Examples of Isotope Notation

$^{14}_{6}C$

$^{28}_{14}Si$

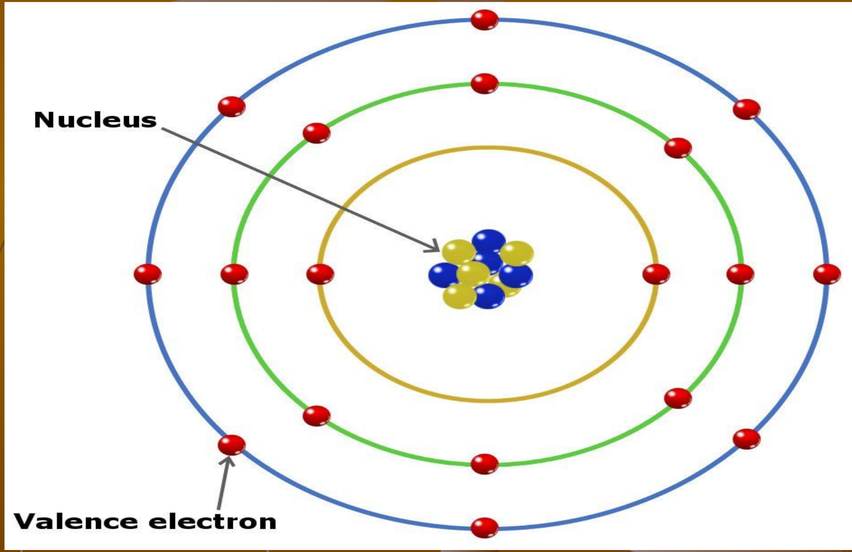
$^{29}_{14}Si$

$^{30}_{14}Si$

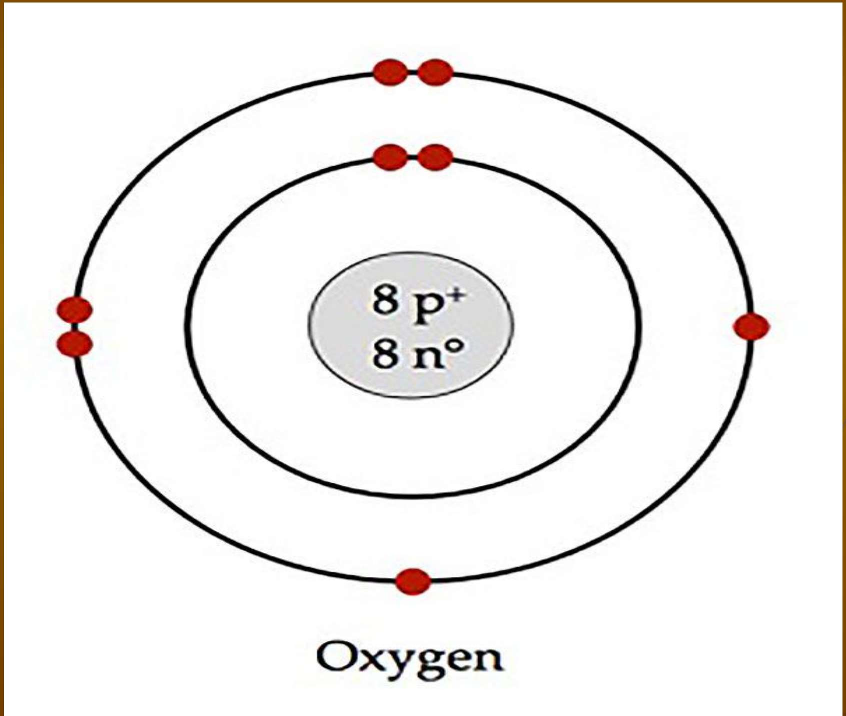


Valence Electrons and Orbits

- Electrons orbit the nucleus in different pathways or electron shells or orbits
- Bohr-Rutherford Model (pictured)
- The outermost orbit or shell is called the valence shell
- The electrons in the valence shell are called valence electrons



Bohr Rutherford Model

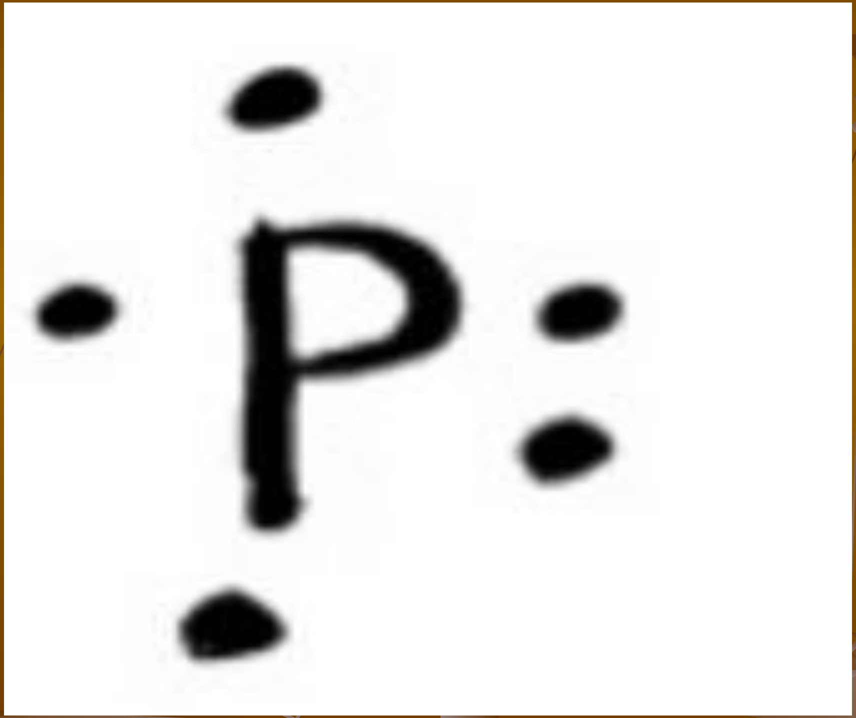


To draw a Bohr Rutherford Model you need:

1. The number of electron shells
2. The number of protons, neutrons, and electrons

Electron Dot (Lewis) Diagrams

- No shells
- Only show the valence electrons and chemical symbol
- Put valence electrons in one at a time; don't match until 4 are in place



Chemical Reactions

- A chemical reaction happens when 2 compounds or elements undergo a **chemical change**

- Recall, a chemical change uses up the starting materials and produces something new

  Example: Baking soda and Vinegar reacts to make water, carbon dioxide and a salt

- What evidence of a chemical change do you see?

Elements vs. Compounds

ELEMENT

- Pure substance
- Made up of the same types of particles
- Most occur in nature (some are synthetic elements)

COMPOUND

- Pure Substance
- Made up of the same types of particles
- May occur in nature
- Formed when 2 or more elements undergo a chemical change making a new substance with unique properties

Compound vs. Molecule

- Molecular compounds can be called compounds and molecules because they are both
- Ionic compounds cannot be called molecules because they are not molecular
- Compounds must contain different elements that can be metals and non-metals or just non-metals (For example: CO_2 and H_2O)

Mixtures vs. Compounds

Mixture

- Impure
- Formed from substances undergoing a physical change
- EXAMPLES:
- Salt water solution, salad, stainless steel

Compound

- Pure
- Formed from elements undergoing a chemical change
- EXAMPLES:
- Sodium Chloride, Lithium Oxide, Carbon Dioxide, Water

Chemical Bonding

- Chemical bonding is the process by which new compounds are made
- A chemical bond is made by lone valence electrons interacting with one another
- There are two types of chemical bonds: Ionic Bonds and Covalent Bonds

Physical Properties of Compounds

Ionic Compounds

- Form as a crystal like shape
- High melting point
- Some are soluble; some are not
- Form electrolytes when dissolved in water
- **Electrolyte:** A liquid substance that conducts a small amount of electricity

Molecular Compounds

- Do not form as crystals
- Low Melting point
- Some are soluble; some are not
- Do not form electrolytes

Ionic Vs. Covalent

- Ionic Bonds hold Ionic Compounds together
- Ionic compounds are made up of a metal element and a non-metal element.
- Electrons from the metal are given to the non-metal because of unequal forces of attraction

- Covalent Bonds hold Molecular Compounds together
- Molecular Compounds are made up of only non-metal elements
- Electrons are not lost or gained, they are shared between elements due to equal forces of attraction

Ionic Compounds

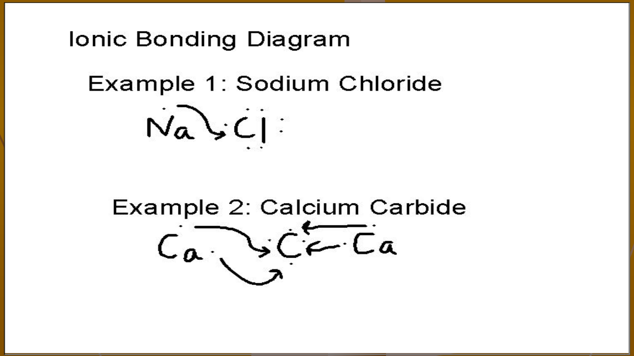
- A metal element and a non-metal element undergo a chemical change
- Salt is an example of an Ionic Compound
- The metal gives **lone valence electrons** to the non-metal because the non-metal has a stronger pull than the metal
- The metal and the non-metal are held together by an **Ionic Bond**

Ionic Bonding Diagram

October 12, 2021

An ionic bonding diagram shows:

1. Two electron dot diagrams; one for each element
2. Arrows to show the movement of lone valence electrons



Nomenclature

- Nomenclature can be defined as the terminology of chemical compounds
- It represents the basic "language of chemistry"
- **Naming Compounds** and **Writing Chemical Formulas**
- How a compound is named depends on what kind of compound it is

Nomenclature for Ionic Compounds

NAME

- The metal is placed first
- The non-metal is placed second and the ending changes to "ide."
- Ex: Sodium Chloride

CHEMICAL FORMULA

- Uses element symbols
- Subscripts are used to show the number of each element in the compound
- Ex: $NaCl$

Ions

- An ion is a charged element. The charges can be used to get chemical formulas for compounds

- Elements become charged when they are ready to bond to another element in ionic bonding due to electron movement

- Metals, non-metals, and metalloids all become charged before Ionic Bonding

Two Types of Ions

CATIONS

- Metal ions
- Positive charge due to imbalance of electrons and protons
- Metalloids on the metal side of the staircase will become cations

ANIONS

- Non-metal ions
- Negative charge due to imbalance of electrons and protons
- Metalloids on the non-metal side of the staircase will become anions

Cation and Anion Symbols

Cation Symbol

The charge is the number of lone valence that are given away in ionic bonding

When negative electrons are given, the ion is more positive

Li^{+}

Mg^{2+}

Anion Symbol

The charge is the number of lone valence electrons that are accepted in ionic bonding

When negative electrons are accepted, the ion is more negative

F^{-}

N^{3-}

Metalloids in Bonding

- In chemical bonding the way a metalloid behaves depends on what side of the staircase it sits on on the Periodic Table
- Those on the metal side act like metals (positive charge, gives electrons)
- Those on the non-metal side act like non-metals (negative charge, accepts electrons)

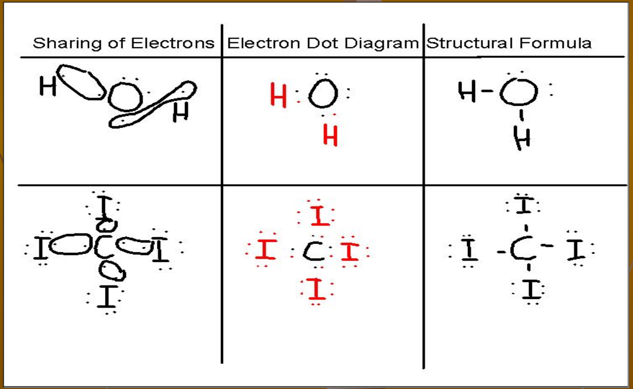
Molecular Compounds

- A non-metal element bonded to another non-metal element
- Water is an example of a molecular compound
- Covalent bonds hold molecular compounds together
- Valence electrons are shared, not lost or gained, because the pull from each element is equal (or close to equal)

Bonding Diagrams for Molecular Compounds

October 12, 2021

- Sharing of Electrons
- Electron dot diagram
- Structural Formula



- The examples are for Water and Carbon Tetraiodide

Nomenclature for Molecular Compounds

- Molecular compounds are named using prefixes.

- For example, CO_2 is called Carbon Dioxide because it is made up of 1 atom of carbon and 2 atoms of oxygen.

- The prefix "di," means 2.

Prefixes

- You should know the following prefixes:
- Mono means 1 atom (mono is not used if there is only one of the first element)

Mono = 1	Tetra = 4
Di = 2	Penta = 5
Tri = 3	Hexa = 6

Try a few…

NAME	**CHEMICAL FORMULA**
Dihydrogen Monoxide	
	SO_2
Carbon Tetrahydride	